

Assignment: Arithmetic and Logic

Instructor: Elaheh Sadredini

elaheh@cs.ucr.edu

University of California, Riverside

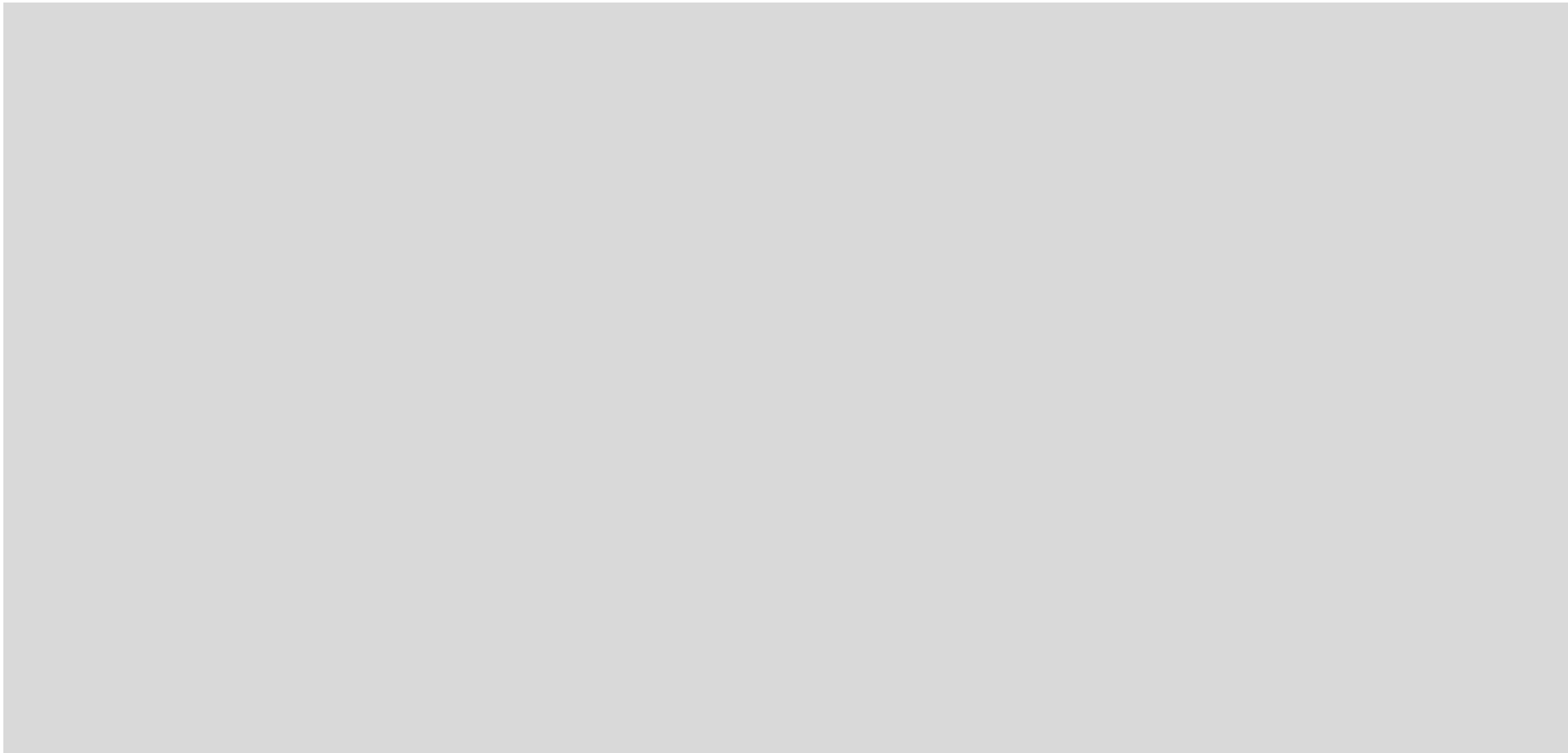
Grading (total 4 points)

- Question 1: 0.4 point
- Question 2: 0.2 point
- Question 3: 1.2 point
- Question 4: 0.6 point
- Question 5: 0.2 point
- Question 6: 1.4 point

Question 1: Two's Complement

Convert the following decimal numbers into 4-bit two's complement:

1. 4
2. -4
3. -8
4. 8



Question 2: Two's Complement Subtraction

Compute the following subtraction in 7-bit two's complement using the

$A - B = A + (!B + 1)$ formula: **8** – **12**

Write the results in binary.

Answer

- Write the final result in binary in 7-bit format

Question 3: Overflow in Two's Complement

Compute if the following operations overflow in 5-bit two's complement. For the answer, provide the carry bits, whether there is an overflow, and the decimal equivalent operation

Hint: for decimal calculation, first convert each number to decimal and then perform the + or – operation.

1. $01000 - 11000$
 - Result and carry-out bit:
 - Does it overflow?
 - Decimal version:

2. $10000 + 01111$
 - Result and carry-out bit:
 - Does it overflow?
 - Decimal version:

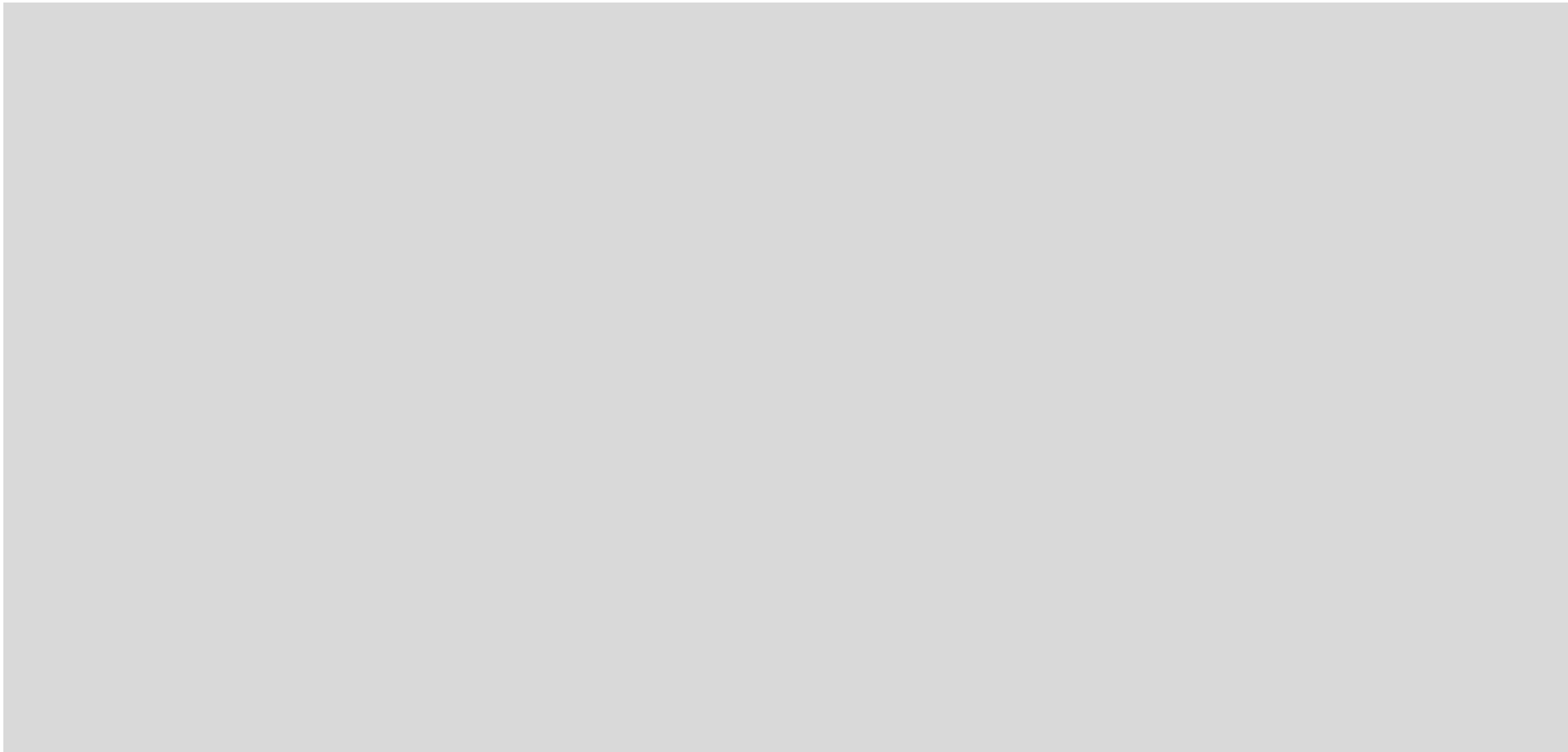
Answer

- Write the final result. For "overflow" question, shortly explain your answer in one sentence.

Question 4: Floating Point

Below is a simplified **8 bit floating point format**, with 1 bit for sign, 4 bits for the fraction, and 3 bits for the exponent. (Format: $Sffffeee = (-1)^S \times 0.ffff \times 2^{(eee-4)}$) For each example, write the decimal equivalent.

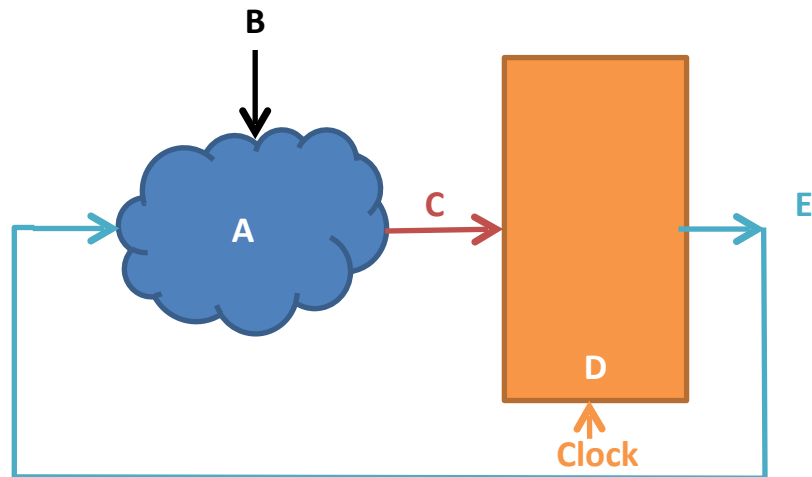
1. 1 0101 110
— Decimal:
2. 0 1111 111
— Decimal:
3. 0 0001 000
— Decimal:



Question 5: State

What do parts A-E need to do in this circuit to make a counter?

1. A: adder, B: input 1, C: next_count, D: state latch, E: current_count
2. A: adder, B: input 1, C: current_count, D: state latch, E: next_count
3. A: adder, B: next_count, C: current_count, D: state latch, E: current_count
4. A: state latch, B: input 1, C: current_count, D: adder, E: next_count



Answer

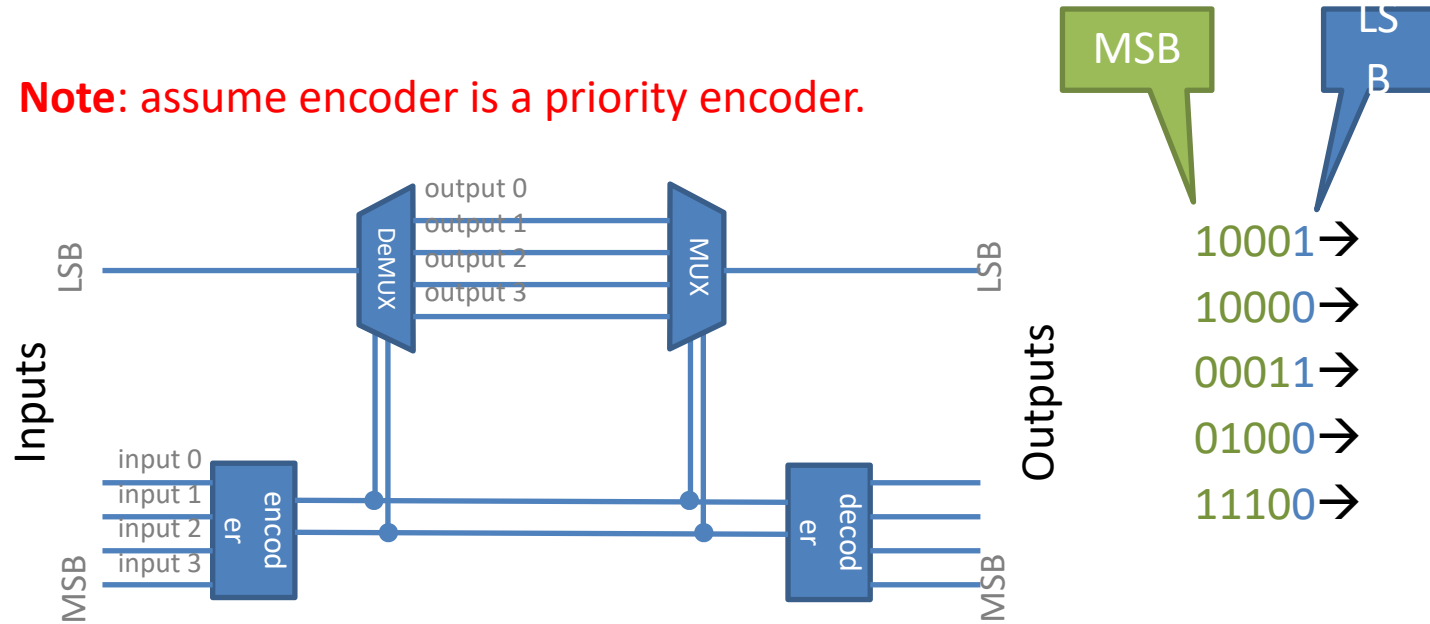
- Choose the correct answer.

Question 6: Logic blocks

A: What is the output for each of the input shown below.

B. Explain your answer for the last one (11100)

Note: assume encoder is a priority encoder.



Answer

- For part A, write the output for each input. For part B, briefly explain your answer.